1 Mark for Review

The ratio of the base to the height of a triangle is 1 to 8. The base of the triangle is 104 centimeters. What is the height, in centimeters, of the triangle?

(A) 13

(B) 52

(C) 104

(D) 832

2 Mark for Review

$$q(x) = 15(3^x)$$

Which table gives three values of x and their corresponding values of $q(x)$ for function q?

A

x	-1	0	1
$q(x)$	-45	0	45

B

x	-1	0	1
$q(x)$	$\frac{1}{5}$	3	45

C

x	-1	0	1
$q(x)$	$\frac{1}{5}$	15	45

D

x	-1	0	1
$q(x)$	5	15	45

3 Mark for Review

A plant nursery classifies a perennial flower species as tall if its typical height when fully grown is more than 100 centimeters. Each of the following inequalities represents the possible heights h, in centimeters, for a specific perennial flower species when fully grown. Which inequality represents the possible heights h, in centimeters, for a tall perennial flower species?

A. $14 < h < 95$

B. $41 < h < 93$

C. $78 < h < 100$

D. $107 < h < 160$

4 Mark for Review

A linear function f gives a company's profit, in dollars, for selling x items. The company's profit is $\$170$ when it sells 4 items, and its profit is $\$590$ when it sells 10 items. Which equation defines f?

(A) $f(x) = 170x - 590$

(B) $f(x) = 59x$

(C) $f(x) = 70x - 10x$

(D) $f(x) = 70x - 110$

5 Mark for Review

Each year, the value of an investment increases by 0.69% of its value the previous year. Which of the following functions best models how the value of the investment changes over time?

(A) Decreasing exponential

(B) Decreasing linear

(C) Increasing exponential

(D) Increasing linear

Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
- You can enter up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer.
- If your answer is a **fraction** that doesn't fit in the provided space, enter the decimal equivalent.
- If your answer is a **decimal** that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), enter it as an improper fraction (7/2) or its decimal equivalent (3.5).
- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	31/2 3 1/2
$\frac{2}{3}$	2/3 .6666 .6667	0.66 .66 0.67

6 Mark for Review

Line p in the xy-plane is defined by $y = x + \frac{8}{3}$ and passes through the points $(a, 0)$ and $(0, b)$, where a and b are constants. What is the value of a?

Answer Preview:

Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
- You can enter up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer.
- If your answer is a **fraction** that doesn't fit in the provided space, enter the decimal equivalent.
- If your answer is a **decimal** that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), enter it as an improper fraction (7/2) or its decimal equivalent (3.5).
- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	31/2 3 1/2
$\frac{2}{3}$	2/3 .6666 .6667	0.66 .66 0.67

$$\frac{25x}{x^2-25} = \frac{x}{x-5} + \frac{20}{x+5}$$

What is the positive solution to the given equation?

Answer Preview:

8 Mark for Review

How many solutions does the equation $7(x+12)+5x=12(x+7)$ have?

(A) Exactly one

(B) Exactly two

(C) Infinitely many

(D) Zero

9 Mark for Review

The manager of a gym selected a sample of 142 members at random to estimate the percentage of the gym's members that would continue to pay for a membership if the price increased. From the survey, the manager estimates that 89% of the gym's members would continue to pay for a membership if the price increased, with an associated margin of error of 5.15%. If the survey is repeated with a random sample of 284 members and the results are calculated in the same way, which of the following will be the most likely effect of using the larger random sample compared to the smaller random sample?

(A) The margin of error will be lower.

(B) The margin of error will be higher.

(C) The estimate of the percentage of the gym's members that would continue to pay for a membership if the price increased will be lower.

(D) The estimate of the percentage of the gym's members that would continue to pay for a membership if the price increased will be higher.

Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
- You can enter up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer.
- If your answer is a **fraction** that doesn't fit in the provided space, enter the decimal equivalent.
- If your answer is a **decimal** that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), enter it as an improper fraction (7/2) or its decimal equivalent (3.5).
- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

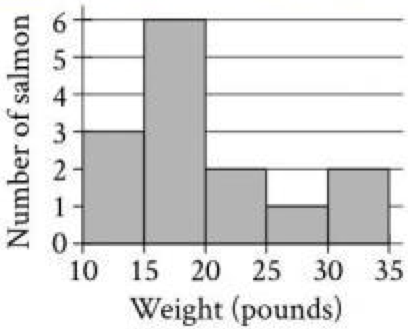
Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	31/2 3 1/2
$\frac{2}{3}$	2/3 .6666 .6667	0.66 .66

10 Mark for Review

The function f is defined by $f(x) = -9x + 9$. What is the y-coordinate of the y-intercept of the graph of $y = f(x)$ in the xy-plane?

Answer Preview:

11 Mark for Review



The histogram summarizes a data set of the weights, in pounds, of 14 salmon. If an additional weight of 68 pounds is added to the original data set to create a new data set of 15 weights of salmon, which of the following measures must be greater for the new data set than for the original data set?

I. The median
II. The mean

(A) I only

(B) II only

(C) I and II

(D) Neither I nor II

12 Mark for Review

x	y
-12	44
a	14
2	b

The table shows three values of x and their corresponding values of y, where a and b are constants. There is a linear relationship between x and y. In the xy-plane, the y-intercept of the line representing this relationship is $(0, -16)$. What is the value of $a + b$?

A. -32

B. -31

C. -22

D. -21

13 Mark for Review

$$(x + k)^2 + 37 = 37$$

In the given equation, k is a constant. How many distinct real solutions does this equation have?

(A) Zero

(B) Exactly one

(C) Exactly two

(D) Infinitely many

14 Mark for Review

For a certain circuit, its power P, in watts; current C, in amperes; voltage V, in volts; and resistance R, in ohms, are related as $\frac{CV^2}{P} = \sqrt{PR}$, where P, C, V, and R are positive. When $R = 18$, which equation correctly expresses P in terms of C and V?

A) $P = \frac{CV^2}{\sqrt{18P}}$

B) $P = \frac{CV^2}{\sqrt{18}}$

C) $P = \sqrt[3]{\frac{C^2V^4}{18}}$

D) $P = \sqrt[3]{\frac{18}{C^2V^4}}$

15 Mark for Review

$$y < -5x - 21$$
$$y > -3x - 8$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given system of inequalities?

A.

x	y
-8	14
-9	35
-11	-3

B.

x	y
-8	19
-9	22
-11	-3

C.

x	y
-8	15
-9	20
-11	28

D.

x	y
-8	17
-9	22
-11	30

16 Mark for Review

A circle is inscribed in a square such that the circumference of the circle touches the midpoint of each side of the square. The length of the diagonal of the square is 228 units. What is the area, in square units, of the circle?

Ⓐ $6{,}498\pi$

Ⓑ $12{,}996\pi$

Ⓒ $25{,}992\pi$

Ⓓ $51{,}984\pi$

Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
- You can enter up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer.
- If your answer is a **fraction** that doesn't fit in the provided space, enter the decimal equivalent.
- If your answer is a **decimal** that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), enter it as an improper fraction (7/2) or its decimal equivalent (3.5).
- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	31/2 3 1/2
$\frac{2}{3}$	2/3 .6666 .6667 0.666	0.66 .66 0.67

17 Mark for Review

The expression kx represents the result of decreasing a positive force, x, acting on an object by 0.34%. What is the value of k?

Answer Preview:

Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
- You can enter up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer.
- If your answer is a **fraction** that doesn't fit in the provided space, enter the decimal equivalent.
- If your answer is a **decimal** that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), enter it as an improper fraction (7/2) or its decimal equivalent (3.5).
- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

18 Mark for Review

A polygon has exactly 87 sides. If the measure of each of the 87 interior angles of this polygon is $(180p)^\circ$, what is the value of p?

Answer Preview:

19 Mark for Review

The number of bacteria in a liquid medium doubles every hour. There are 66,000 bacteria in the liquid medium at the start of an observation. Which of the following represents the number of bacteria, y, in the liquid medium t hours after the start of the observation?

(A) $y = \frac{1}{2}(66{,}000)^t$

(B) $y = 2(66{,}000)^t$

(C) $y = 66{,}000\left(\frac{1}{2}\right)^t$

(D) $y = 66{,}000(2)^t$

20 Mark for Review

The function $f(n) = 7(20.44)^{\frac{n}{4}}$ gives each term of a sequence as a function of the term's position, n, in the sequence, where n is a whole number. The value of the term in position 14 is $p\%$ more than the value of the term in position 10. What is the value of p?

A) 20.44

B) 44

C) $1{,}944$

D) $2{,}044$

21 Mark for Review

A right square pyramid has a total surface area of $28,160$ square inches, and the combined surface area of the four lateral faces of this pyramid is $16,060$ square inches. What is the height, in inches, of this pyramid?

A) 48

B) 55

C) 73

D) 110

Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
- You can enter up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer.
- If your answer is a **fraction** that doesn't fit in the provided space, enter the decimal equivalent.
- If your answer is a **decimal** that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), enter it as an improper fraction (7/2) or its decimal equivalent (3.5).
- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

22 Mark for Review

The expression $6x^4 + 17x^2 + 7$ can be rewritten as $(3x^2 + a)(2x^2 + b)$, where a and b are positive integers, or as $(3x^2 + c)(2x^2 + d)$, where c and d are positive nonintegers. What is the value of $a + c$?

Answer Preview: